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The Implications of Automated Production Solution on both Production Optimization & Subsurface Water Management, Mature Asset in Libya

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Abstract

Objectives/Scope: The upstream oil and gas industry is experiencing a paradigm shift, driven by the imperative to enhance reservoir performance through digital innovation. Within this context, Digital Oilfield 4.0 (DOF 4.0) represents a transformative advancement in reservoir management, integrating real-time data acquisition, advanced analytics, and intelligent automation to optimize production, reduce non-productive time (NPT), and improve asset profitability.

Methods, Procedures, Process: This paper presents the successful deployment of Automated Field Production Solution, a unified digital platform developed under the DOF 4.0 framework and implemented at Sarir Oil Operations B.V. Libya Branch in Q3 2023. Automated Field Production Solution merges surveillance, diagnostics, and optimization workflows into a single, configurable environment, allowing seamless cross-disciplinary collaboration and data-driven decision-making. The platform is architected around the 4R Approach—delivering the right data to the right person at the right time to make the right decision—thus ensuring operational agility and responsiveness.

Results, Observations, Conclusions: Key functionalities include Surveillance by exception-based monitoring, virtual flow metering, gas lift optimization, and automated forecasting tools, all accessible through a web-based interface. These combined workflows have significantly enhanced the efficiency of daily operations and standardized engineering practices across the asset.

In parallel, Automated Field Production Solution supported the strategic resumption rigless workover activities, which were leveraged not only for production enhancement but also as platforms for experiential learning and team development. This dual-purpose approach resulted in a net production gain exceeding 30% & reduced produced water with 15% in 2024, while simultaneously cultivating a technically proficient and autonomous reservoir engineering team.

The integration of Automated Field Production Solution within the DOF 4.0 ecosystem has delivered significant improvements in operational efficiencies, data transparency, and organizational capability.

Novel/Additive Information: This paper demonstrates how the convergence of digital field surveillance, intelligent workflows, and structured competency development can unlock sustainable value in mature assets and serve as a blueprint for future digital transformation initiatives in reservoir management.

Introduction

Automated field production solution technology is an industry unique full-field surveillance and optimization framework, and established workflow library designed to automate broad, upstream engineering processes. The Workflow Library has grown to address a very broad portion of the processes typical of upstream engineering and operations.

The implementations start by integrating siloed and disparate data (see Figure 1). The premise behind the digital solution is to enable the virtual integration of data from segmented systems/databases/applications/excels so that data/information appears to come from a single source. Any connected data source is automatically referenced by Digital solution, so the latest data is always available to the Workflows. Clients routinely report that this integration reduces the time spent by engineers looking for data by 75%.

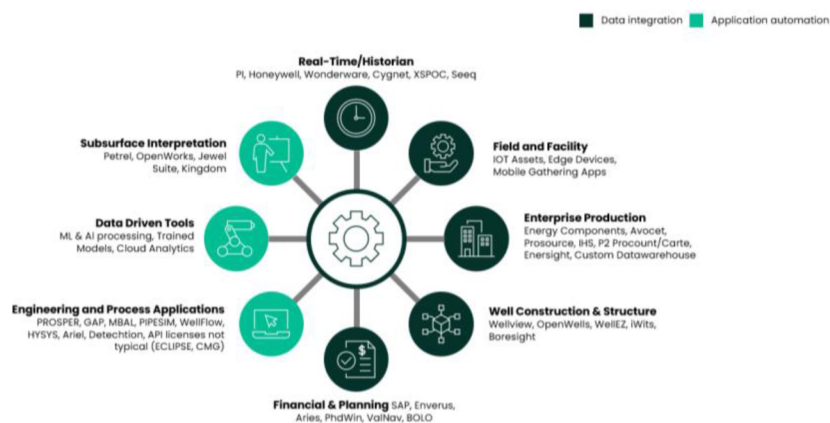


Figure 1—Technology Data Integration

Once the data is virtually connected, the digital solutions workflows consume the data to support broad engineering processes: automated well reviews, data QC, well-opportunity ranking, well opportunity management system, surveillance by exception, task management (capture, assign, and track), and integration of complex third-party physics or data-driven modeling tools, providing a complete suite of workflows to support surveillance, opportunity management, forecasting and optimization.

Methods, Procedures, Process

Well Performance Analysis (WPA)

WPA is used to surveil natural and gas lift wells, including the integration of third-party well model applications, e.g., PROSPER, PIPESIM. The Well Performance Analysis (WPA) module is a collection of workflows to help optimize production at the well level. WPA is configured for naturally flowing, ESP, and gas lifted wells.

It uses all the well data available and leverages well models as part of the workflow.

Well Performance Analysis is composed of 4 primary workflows: Well Test/Model Validation with user evaluation, Virtual Metering (VM), Diagnostics, & Lift Optimization.

Well Test/Model Calibration

The workflow integrates data recorded in the field or as part of well tests and automatically compares the results to the current well-model results. The data used for this validation may be taken from real-time data or other sources within the client's architecture. Validation workflows may be triggered on a predetermined Schedule, such as nightly or hourly, or triggered by an event such as a surveillance alert or a well test entered into the system. Well models are calibrated in a single-model approach and can then be incorporated into a network calibration. When the validation workflow is completed, it will generate comparisons between "real-life" and the model. Many different parameters may be used in the comparison. Common parameters are produced volumes or bottom-hole pressures. If there is a significant error or difference, the workflow may trigger an alert or task in the opportunity management system. The threshold for what constitutes a "significant error" is a configurable setting that can be set at deployment and modified at any time. The alert may suggest that the model needs to be updated or perhaps that there may be some error in the well test. [Figure 2](#).

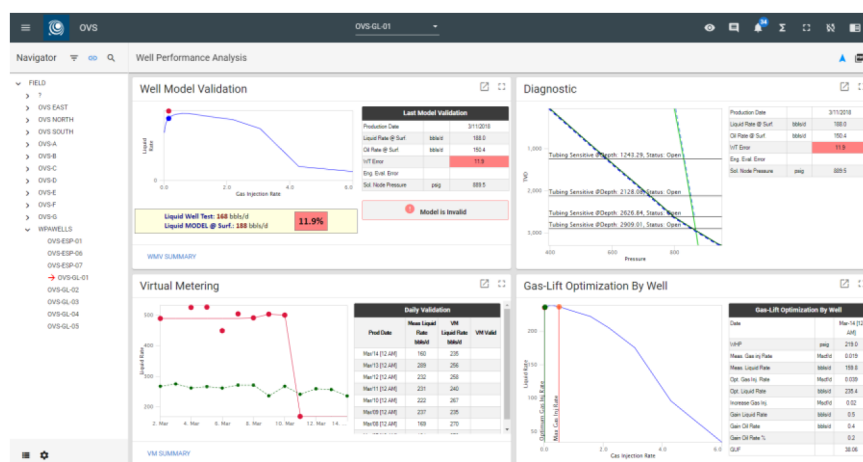


Figure 2—WPA – Well Performance Dashboard (Gas-Lift Example)

The automated technology Surveillance by Exception (SBE) tool can be used to alert the operations team when current operating points are close or outside of the safe operating envelope. Other key performance indicators can also be monitored and generate alerts using the SBE.

This workflow includes a quick "data validation" to provide an insight to the engineers if the well test data is trustable or requires attention from the engineer. The validations include :

- Validate that the operational parameters, flowing conditions are somewhat stable between the last and previous well tests, key variables like ESP frequency, gas lift injection rate, well head pressure are monitored looking for relevant variations, in such cases, the workflow will automatically show an alert for validation.
- If not, considerable changes occurred in the well's operational conditions, then the well test should follow a trend with an auto decline tool within certain limits. The validation is automatically run and show as well as an alert to the engineers.

This alert will be shown as part of the Workflow 2, but the engineers can "click" and make their own assessment. [Figure 3](#) shows how these workflows are linked together.



In cases where the GOR values may have uncertainties, the Well Model Validation workflow includes a configurable option, that can be enabled/disabled, that allows to run a pre-calibration process of the GOR by using the Choke Performance module in PROSPER. Operational data like THP and LP (used as inlet/outlet pressure), water cut, choke size, WHT is used as input, and the process iterates until it finds a GOR that matches the production rate obtained from the well test. The obtained GOR can then be set in the model and the complete well model validation process is run. For Gas Lifted wells, the calibrated GOR can be used to estimate the gas lift injection rate, using a formation GOR as reference, for example the Rs defined in the PVT properties. [Figure 4](#) shows a case when this part of the workflows is activated, automatically calculates the estimated Gas Lift Rate and compares it with the actual Gas Lift Rate reported.



Diagnostics

Using the same models, the WPA tool will automatically use the current operating conditions to run certain diagnostics on the well. For gas-lifted wells this will generate a determination of the pressure gradients in the well and determine the current operating valve as shown in the upper right section of [Figure 2](#).

Virtual Metering

A calibrated and accurate model can be used to predict the production rates for a well. The current operating parameters are automatically input into the model and production values are calculated. These virtual rates can be used to estimate production between well tests for comparison to sensor data or when sensor data is not available. These values aid in production allocation, flow assurance, failure prediction, and short-term forecasting.

The Well Performance Analysis tool provides a mechanism for performing regular virtual metering assessments. The tool is scheduled to execute on a timing that makes sense for each asset's unique operating conditions. It is common to perform virtual metering daily, but the schedule can be set to run with shorter intervals. Figure 5 is an example of the results of a virtual-metering run with the history of the previous evaluations.

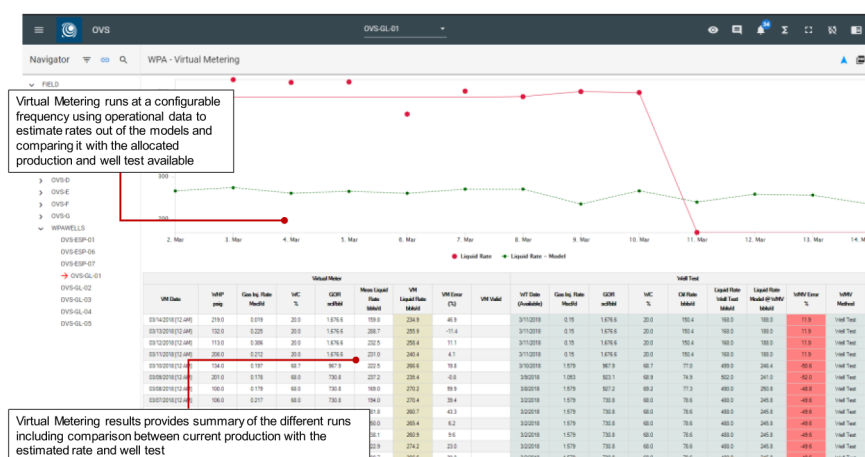


Figure 5—WPA – Virtual Metering Summary.

Virtual Metering for Real-Time Surveillance

Another important application of this workflow is on real-time surveillance, the calculated rate can be used to validate the well is operating at desirable condition, within a specific operating envelope or user provided limits. Wells producing via ESPs are surveilled by Leucipa using the pump operating envelopes, validating if the well operational conditions are close or over of the upthrust/downthrust limits, generating alerts to prevent additional wear and tear of the pump.

For Gas Lift or Natural Flowing wells, the system allows the engineers to provide operating limits configurable per well, area, reservoir, etc. limits like: BHP operation (BHP calculated by the model, assuming that DHG are not available), WHP limits, Gas Lift Rate (range), among others.

Gas Lift Optimization

Automated field production technology created a mathematical approach that calculates the optimum Gas Lift, using the gas lift performance curve from well model software. This is not the maximum value in the performance curve. Instead, Leucipa uses the concept of Gas Utilization Factor (GUF) which indicates how much gas is needed to lift a barrel of oil. Using a configurable factor and current well conditions, the

technology will indicate at what point along the performance curve the well should be operating in order to be at the optimum value.

When this is performed for all wells, the entire platform or field can be selected, and a summary populates of the opportunities to increase production with the new set points. This workflow will not be accurate if the Well Model Validation is not adjusted. For that reason, automated field production technology shows an alarm from this workflow indicating whether you can trust this incremental production prediction (tied to the model validation).

Final gas lift optimization (GLO) should be done at the network level where all the pressure drops and constraints are considered for Field Optimization. Figure 6 shows an example of the Well Performance Analysis dashboard with a gas lift optimization summary graph.

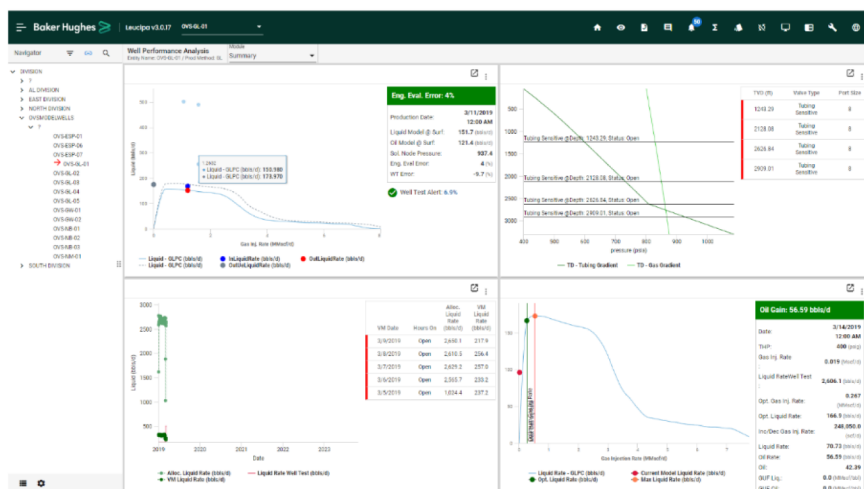


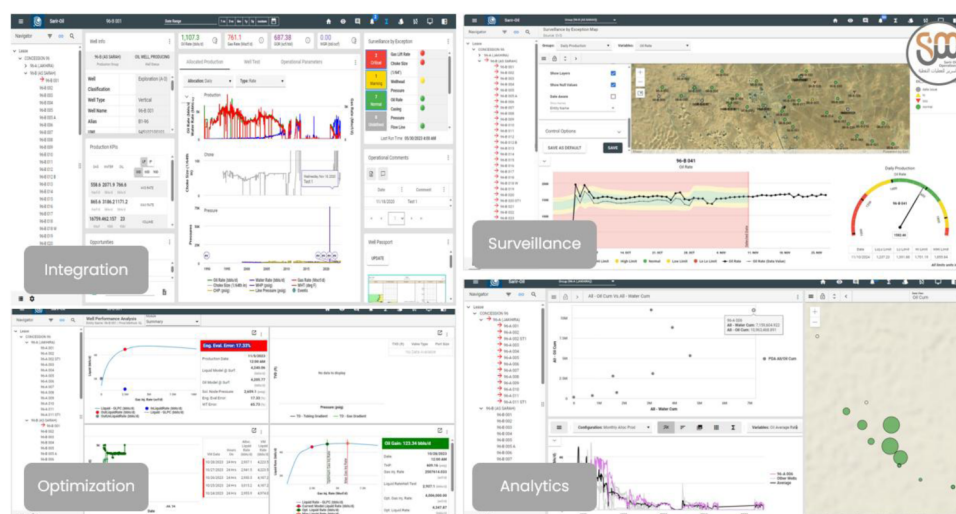
Figure 6—Well Performance Analysis Summary Including Gas Lift Optimization

Results, Observations, Conclusions

Classic Digital Oilfield combined with Automated field production Technology significantly advanced reservoir management by automating data analytics, surveillance, and production optimization. This transformation enables Real-time data collection and analysis leading to more accurate decision-making & Optimized production processes

Figure 7 shows the Integrating of these elements enhances operational efficiency, reduces downtime, and improves collaboration across various disciplines. Achieving complete success relies on users mastering the technology and applying the best practices fully.

By implementing such a system, Engineering and asset management teams are equipped to realize the industry-wide push toward digital oil field, data connectivity, and integrated operations with a single solution that providing data integration, visualization, continuous surveillance, and workflow automation tools. Enabling more effective analysis & making a rightful decision at the right time



Deployed certain workflows such as WDB, MWDB, SBE, WPA to optimize the production. Relied on model management to keep all models evergreen automatically. Run different model-based operational scenarios to increase production. Test model scenarios on actual fields to establish level of success. The automation process results in more of value performance results as in [table 1](#) below.

Table 1—Key Performance Results from Automated Production Solution Deployment

PARAMETER OR KPI'S	IMPACT
FIELD PRODUCTIVITY	37% increase
WATER CUT	25 % reduce
LIFTING COST	15 % dropped
GAS FLARING	zero
HSE	6 years LTI free

Nomenclature

- DOF:** Digital Oil field
- NPT:** Non productive time
- VM:** Virtual metering
- PVT:** Pressure-volume-temperature
- LP:** Lifting pressure
- THP:** Tubing Head pressure
- GOR:** Gas Oil Ratio
- WHT:** Wellhead Temperature
- WDB:** Well data book
- MWDB:** Multi-well data book
- SBE:** Surveillance by exception
- WPA:** Well Performance Analysis
- GLO:** Gas lift Optimization
- GUF:** Gas utilization factor
- ESP:** Electrical submersible pump
- WHP:** Well head pressure
- DGP:** Downhole gas pressure

BHP: bottomhole pressure

References

- El Sherbeny, W. et al; *The Power of Digital Oilfield 4.0 | Transforming Reservoir Management*. SONATRACH Scientific and Technical Days. Oran, 24-26 June 2025.
- El Sherbeny, W. et al; SPE-221571-MS: An Artificial intelligent cutting-edge technology for automated field production optimization & ESP Failures prediction. Middle East artificial lift conference and exhibition, Bahrain, 2024.
- Sweden, A. et al; *Transforming Reservoir Management; The Power of Digital Surveillance 4.0*, TOGSE, 2024
- Sweden, A. et al: FIELD PERFORMANCE IMPROVEMENTS THROUGH INTEGRATION AND AUTOMATION, Baker Hughes AM, 2025
- El Sherbeny, W. et al; Automated Field Production Optimization & ESP Failures Prediction; Technology Overview and Global Case Histories; SPE Kuwait international symposium 27-28th November 2024.
- El Sherbeny, W. et al: MLF79: An Artificial intelligent cutting-edge technology for automated field production optimization & ESP Failures prediction, 10th Middle East artificial lift forum, Oman, 2025.